

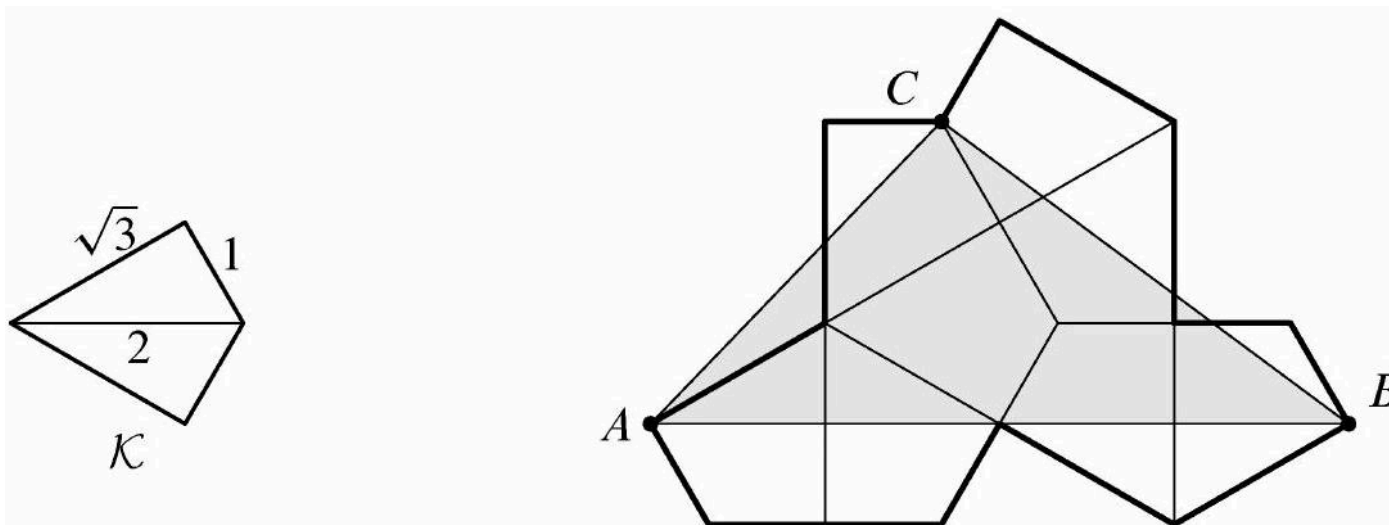
2024 AMC 10A Problem 22 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 22. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 22

Problem:

Let $\mathcal{K}$ be the kite formed by joining two right triangles with legs 1 and $\sqrt{3}$ along a common hypotenuse. Eight copies of $\mathcal{K}$ are used to form the polygon shown below. What is the area of triangle $\triangle ABC$?



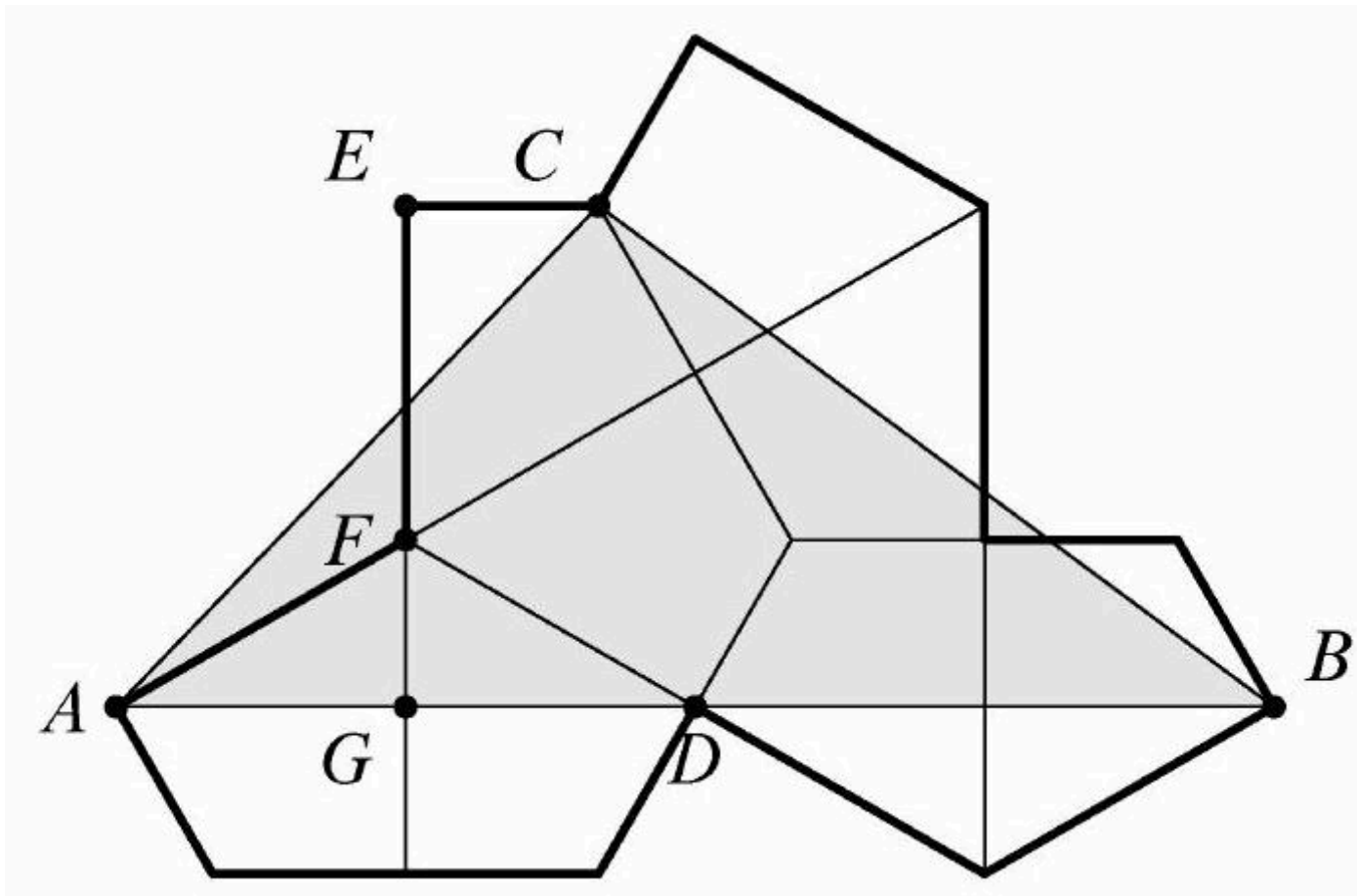
Answer Choices:

- A. $2 + 3\sqrt{3}$
- B. $\frac{9}{2}\sqrt{3}$
- C. $\frac{10+8\sqrt{3}}{3}$

D. 8

E. $5\sqrt{3}$ **Solution:**

Let points D , E , F , and G be labeled as shown. Points A , D , and B are collinear, and the line through them is perpendicular to line EF . Furthermore, the distance from C to line AB is equal to the distance from E to line AB , which in turn is equal to EG . The area of $\triangle ABC$ is thus $\frac{1}{2} \cdot AB \cdot EG$.



To compute AB , observe that $\overline{AG}$ is the longer leg of a $30^\circ - 60^\circ - 90^\circ$ triangle with hypotenuse $\sqrt{3}$. This implies that $AG = \frac{3}{2}$, from which $AB = 4 \cdot AG = 6$. To compute EG , observe that in the same right triangle as before, $FG = \frac{\sqrt{3}}{2}$. This implies that $EG = EF + FG = \sqrt{3} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$.

Therefore the area of $\triangle ABC$ is $\frac{1}{2} \cdot 6 \cdot \frac{3\sqrt{3}}{2} = \text{(B)} \boxed{\frac{9\sqrt{3}}{2}}$.

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗

Powered by [Wiki.js](#)